IoT Homework 2025/2026

Integrated Report: RFID Dynamic Frame ALOHA

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Exercise 3 - RFID Dynamic Frame ALOHA

A Dynamic Frame Slotted ALOHA reader must identify N = 4 tags. The initial frame size r1 is varied from 1 to 6. After the first frame, the frame size is correctly set to the current backlog size. The efficiency is eta(r1) = N / E[L(r1)], where E[L(r1)] is the expected number of slots needed to identify all tags.

3.1 Matched-frame resolution costs

The values L2 = 4 and L3 = 51/8 are given. For N = 4, the matched-frame expected cost L4 is obtained by distributing four tags into four slots. The possible remaining backlogs are 0, 2, 3, and 4, with counts 24, 144, 48, and 40 over 4^4 = 256 equally likely outcomes.

Thus:

L4 = 4 + (144/256)L2 + (48/256)L3 + (40/256)L4.

Solving gives L4 = 953/108 = 8.8241 slots.

3.2 Efficiency for r1 = 1, 2, 3, 4, 5, 6

For a first frame of size r1, the number of outcomes is r1^4. The remaining backlog counts after the first frame are:

- b = 0: N0 = r1(r1-1)(r1-2)(r1-3)
- b = 2: N2 = 6r1(r1-1)(r1-2)
- b = 3: N3 = 4r1(r1-1)
- b = 4: N4 = r1 + 3r1(r1-1)

The expected total resolution length is:

E[L(r1)] = r1 + (N2/r1^4)L2 + (N3/r1^4)L3 + (N4/r1^4)L4.

Initial frame size r1	E[L(r1)] slots	Efficiency eta = 4/E[L]
1	9.8241	0.4072
2	9.5995	0.4167
3	8.9544	0.4467
4	8.8241	0.4533
5	9.0377	0.4426
6	9.4661	0.4226

3.3 Efficiency plot and maximum

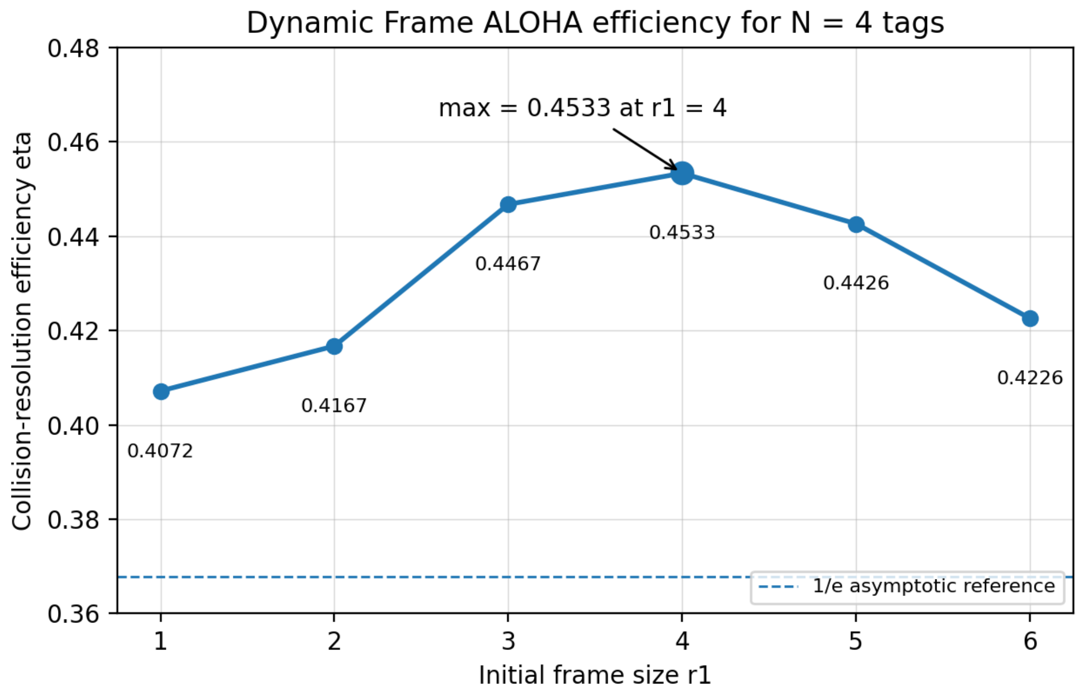

Figure 1. Collision-resolution efficiency as a function of the initial frame size r1 for N = 4 tags.

The maximum efficiency is obtained at r1 = 4 = N, with eta_max = 0.4533. This is expected: choosing an initial frame size close to the number of tags maximizes the probability of singleton slots while limiting both empty slots and collisions. Smaller r1 values create too many collisions, while larger r1 values waste capacity through empty slots.

Final Numerical Values for Form Entry

Quantity	Value
Exercise 3 - eta values for r1=1..6	0.4072, 0.4167, 0.4467, 0.4533, 0.4426, 0.4226
Exercise 3 - maximum eta	0.4533 at r1 = 4